MKO · DATA · IA · SOFTWARE



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LANGUAGES

Français · native

English · professional (B2)

CERTIFICATIONS

Huawei · Artificial Intelligence Technology and Application V2.0

Meta · Front-End Developer (HTML, CSS, JavaScript, React)

Board Infinity · .NET Developer & ASP.NET Core

INVOLVEMENT & INTERESTS

ENSPD AI Club · head of R&D (2024)

Competitions · Zindi, Data Tour

Interests · music, reading, cinema, cooking, football

CURRICULUM VITAE

Georges Miguel MOUKOKO EKAMBI

AI Engineer & Data Scientist

Explainable AI · Edge AI · Full-stack · Founder of LIDIA Lab · MSc Data Science & AI, ENSPD Douala

> Final-year MSc Data Science & AI engineer. I design explainable models and ship them to production. My two research tracks are explainable AI (Kolmogorov-Arnold networks, SHAP, conformal prediction) and edge AI (quantised models that run offline). I have just completed a Data Scientist internship at the BEAC, where I delivered an anomaly detection application and led a balance of payments study. Founder of LIDIA, a sovereign-AI startup, and software development instructor at ENSPD.

01 EXPERIENCE

Founder & AI Lead

2025 · present

LIDIA · sovereign AI startup

- Hybrid RAG architecture (semantic + BM25 retrieval) and multimodal OCR pipelines for closed intranet deployment.
- Products: AgroSense, Nexus AI, Meerkat, PharmaCon and Lead's AI, a sovereign LLM adapted to African languages.
- Applied research in explainable AI and edge AI.

Software developer

2026 · present

NexCQISolutions · Aura CareOS

- Security module of an OpenEMR-based electronic medical record: MFA, access control, tamper-evident audit log, PHPUnit tests.

Data Scientist Intern

08/2026 · 10/2026

BEAC (Bank of Central African States) · Yaoundé · Research and Statistics Directorate (DERSRI)

- Main assignment in the Monetary Statistics and Analysis Department: delivered an anomaly detection application for the bank balance sheets of the six CEMAC countries (12 files, 86 monthly IFS indicators), with no labelled data.
- Compared three models: LOF*, a Kolmogorov-Arnold network (MKAN) and BiVAT (Transformer + BiLSTM + VAE) with conformal prediction and SHAP. BiVAT isolated the COVID-19 shock on its own. Dash + PyQt6 desktop software with roles and PDF reports.
- Balance of payments: 2021 diagnosis of Congo, Gabon and Chad, then export value retention across 41 oil exporters over six years (IMF, BEAC, BPM6): five-band typology, transitions, leakage profiles with k-means and PCA.
- Oil and gas sector outlook for CEMAC, industrial benchmark of the six states and rotation across nine departments of the directorate.

Software engineering intern

07/2025 · 01/2026

ESOAF · Yaoundé · Full-stack and AI

- Designed and built full-stack modules for enterprise data anonymisation.
- Integrated computer vision models into industrial workflows.
- Structured data flows and processing pipelines for decision support.

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SKILLS

AI & Data Science
Python, R, PyTorch, TensorFlow / TFLite, scikit-learn, XGBoost, SHAP, Optuna, time series, anomaly detection, RL

Explainable & edge AI
KAN, Grad-CAM, conformal prediction, INT8 quantisation, YOLOv8, EfficientNet, offline-first

Generative AI & agents
Hybrid RAG (BM25 + dense, RRF), GraphRAG, LangGraph, Qdrant, Ollama, multimodal OCR, GAN, VAE

Economics & finance
monetary statistics (IFS), balance of payments (BPM6), economic outlook analysis, PCA, k-means

Engineering
C# / ASP.NET Core 8, FastAPI, Node.js, React, TypeScript, Flutter, PyQt6, Rust, PHP, Clean Architecture

Data & Ops
PostgreSQL / PostGIS, SQLite, Redis, Kafka, Linux, Docker, AWS EC2 / S3, GitHub Actions, Caddy, Plotly Dash

SCAN ME

PORTFOLIO LINKEDIN GITHUB

01 EXPERIENCE

Full-stack developer and software development instructor
Alongside

Freelance · ENSPD

- WinPlus (live learning platform), multi-tenant business email, HomeGamesFive, SISIA SARL.
- Mobile Money payment integration (MTN MoMo, Orange Money) and AWS deployment.
- Software development instructor at ENSPD.

Intern
2022 · 2023 · 2024

CAMRAIL · Douala · three summer internships

- Supported railway traffic management teams.
- Wrote technical reports on operational performance indicators.
- Contributed to transit-time optimisation projects.

02 EDUCATION

MSc Data Science & AI (SDIA) — Ongoing
ENSPD, University of Douala · thesis: Mobile Money fraud detection with Kolmogorov-Arnold networks

BSc Data Science & AI — 2024
ENSPD · with honours

Baccalaureate, series C (maths & physics) — 2021
Collège Adventiste de Yaoundé · with honours

03 KEY PROJECTS

MKAN · MoMTSim
Thesis · explainable AI
Mobile Money fraud detection with a recurrent KAN, multi-agent simulator, explained real-time scoring (Kafka, Redis, Dash).

BEAC Detect
BEAC · anomaly detection
LOF*, MKAN and BiVAT on CEMAC bank balance sheets, conformal prediction, SHAP; Dash + PyQt6 software delivered.

Export value retention
BEAC · balance of payments
41 oil exporters, 246 observations: five-band typology, transitions, leakage profiles (k-means, PCA).

KALATI-RAG
CAMRAIL · sovereign RAG
On-premise document assistant: hybrid retrieval, GraphRAG, faithfulness gate, Qwen3 via Ollama.

PREDOC
CAMRAIL · forecasting
Block occupation time after a derailment: quantile XGBoost, calibrated interval, SHAP.

AgroSense
LIDIA · edge AI
Offline photo-based crop diagnosis (TFLite), Grad-CAM, DBSCAN surveillance; Flutter, .NET, FastAPI.